## Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

### 2 Outline three benefits of 3D printing when producing products.

- No tooling or moulds needed
- Complex forms and moving parts in one print
- Prints directly from a CAD file
- Easy to change the design and reprint
- Additive, so little waste
- Good for one-off or customised parts

**No marks for:** Unqualified "quick", "cheap", "easy"; Vague environmental claims

### 3 Quick check

- a) High initial investment and training
- b) Acrylic earrings with intricate cut-outs
- c) No moulds are needed so changes are cheap and quick
- d) Brown scorch marks

### 4 Key terms

- 1 – D
- 2 – C
- 3 – A
- 4 – B

### 5 Describe one benefit of 3D printing.

- Complex forms with no tooling; easy to change; low waste.

### 6 Describe one drawback of 3D printing.

- Slow and limited materials; not economic for high volumes.